



DTE Code: EN4139

Samridhi Sarwajanik Charitable Trust's
JHULELAL INSTITUTE OF TECHNOLOGY

An Autonomous Institute affiliated to RTM Nagpur University

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

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Vision: To evolve as the premier department in the field of AI & ML through Enriched
Learning environment, Innovation and research to cater industry-socio needs.



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Synopsis
on
KRYNTRA
Continuous & autonomous agentic cybersecurity assessment platform

A synopsis submitted in Partial Fulfillment of the requirements for the Degree of
BACHELOR OF TECHNOLOGY

in
COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine learning)

(III Semester)

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DEPARTMENT VISION

To evolve as the premier department in the field of AI & ML through Enriched Learning environment, innovation and research to cater industry-socio needs.

DEPARTMENT MISSION

1. To enhance employability through skill development, research and industry-institute interaction.
2. To develop professionals with ethical values and social.

Department PEO's

- 1) To develop an ability to analyze the software, understand the technical specifications, design and provide novel engineering solutions and efficient product designs.
- 2) To develop professional skills that prepares them for immediate employment and for life-long learning in advanced areas of computer science and related fields.
- 3) To develop necessary communication skills to bridge the gap between advanced technology and end users in practice of computer science products.
- 4) To develop technical skills to adapt to an ever-changing professional environment.



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Title:

KRYNTRA : Continuous & Autonomous Agentic Cybersecurity Assessment Platform

Aim:

The primary aim of KRYNTRA is to develop a continuous and autonomous cybersecurity assessment platform capable of identifying security weaknesses, intelligently analyzing vulnerability findings, and supporting safe remediation. The system is intended to replace highly manual and infrequent security assessment practices with a continuous workflow that combines automated security tools, AI-driven decision support, real-time monitoring, and sandbox-based verification.



Introduction:

- Cybersecurity has become a continuous requirement because new vulnerabilities can emerge at any time. Traditional security audits are often manual, expensive, infrequent, and disruptive. This creates a security gap: a system may be assessed at one point in time, while a new vulnerability discovered afterwards can remain unnoticed until the next assessment cycle. Small and Medium Enterprises (SMEs) can face an additional challenge because maintaining a dedicated security operations team may not be practical.
- KRYNTRA is proposed to bridge this gap through an autonomous, agentic cybersecurity workflow. It combines raw scanning capabilities with AI-based intelligence so that security findings can be collected, normalized, analyzed, prioritized, and acted upon through a single platform. The platform performs continuous asset discovery and uses Nmap, OWASP ZAP, and Trivy for multi-layered assessment of networks, web applications, containers, and filesystems.
- The collected scan results are converted into a unified schema and passed to an AI triage engine using LangChain and an LLM. The AI layer analyzes the context of findings, helps filter false positives, prioritizes vulnerabilities by severity, and maps findings to the OWASP Top 10 framework. For remediation, the platform can generate proposed code patches or security changes. These proposed fixes are tested inside an isolated Docker-based sandbox before deployment, reducing the risk of applying an unverified change directly to a production environment.
- A real-time dashboard provides visibility into active scans, threat severity, vulnerability logs, AI-generated remediation suggestions, and sandbox verification results. The platform also includes compliance-oriented assessment and reporting capabilities for SOC2 and ISO27001 readiness. Thus, KRYNTRA brings together vulnerability discovery, AI-driven triage, remediation assistance, verification, monitoring, and reporting into one continuous security assessment workflow.



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Objective:

- Automate continuous discovery, profiling, and tracking of network and cloud assets.
- Perform multi-layered vulnerability assessment using Nmap, OWASP ZAP, and Trivy.
- Normalize raw scanner outputs into a unified format suitable for centralized analysis.
- Use AI/LLM-based triage to analyze security findings, reduce false-positive noise, and prioritize vulnerabilities according to severity and context.
- Map relevant vulnerability findings to the OWASP Top 10 framework.
- Generate automated remediation suggestions, including proposed code patches and security configuration changes.
- Execute proposed fixes in an isolated sandbox to verify their effectiveness before deployment.
- Provide a real-time dashboard with scan progress, security metrics, vulnerability logs, and remediation status.
- Generate compliance-readiness information and audit-oriented reports related to SOC2 and ISO27001.



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Scope of problem:

- The scope of KRYNTRA covers the automation of major stages of a cybersecurity assessment workflow. The proposed system begins with asset discovery and proceeds through network, web application, container, and filesystem vulnerability scanning. It then processes scanner output through an AI triage layer, prioritizes findings, provides remediation assistance, verifies proposed fixes in a safe environment, and presents the results through a centralized dashboard.
- The system architecture is designed around a modern distributed workflow. The frontend uses Next.js 14 with Tailwind CSS and Shadcn UI, while FastAPI provides the backend API layer. Celery and Redis handle long-running scanning tasks asynchronously. SQLite/PostgreSQL can be used for persistence. Nmap, OWASP ZAP, and Trivy operate as scanning components, while LangChain and an LLM support intelligent triage. Docker/Docker Compose support containerization and sandbox environments.
- The future scope described for KRYNTRA includes integration with cloud environments such as AWS, GCP, and Azure; customized AI agents with additional security tools such as Metasploit and Nuclei; and automated deployment through GitHub Actions and CI/CD integration.



Reasons for Selecting the Problem:

- The problem was selected because cybersecurity assessment cannot be treated as a one-time activity. Vulnerabilities and threats can emerge continuously, creating exposure between conventional audit cycles. Manual assessment also requires considerable time and skilled human effort, while security teams may receive large volumes of raw findings that contain noise and potential false positives.
- KRYNTRA addresses these practical challenges by combining automated scanning with AI-assisted interpretation and safe verification. The project is therefore relevant to modern cybersecurity requirements and also provides an opportunity to apply Artificial Intelligence and Machine Learning concepts to a real-world engineering problem. Its focus on SMEs is particularly important because it aims to make continuous security assessment more accessible without relying entirely on large dedicated security teams.

Theoretical and Practical Aspects:

The theoretical aspect of the project involves cybersecurity assessment, vulnerability management, network and application security, AI/LLM-based reasoning, asynchronous task processing, distributed architecture, and containerized execution. The practical aspect involves integrating established security scanners with an AI-driven workflow, building a real-time dashboard, storing and processing vulnerability data, generating remediation suggestions, and testing proposed fixes inside an isolated sandbox. This combination demonstrates how software engineering and AI techniques can be applied to improve cybersecurity operations.



Proposed Plan of Work:

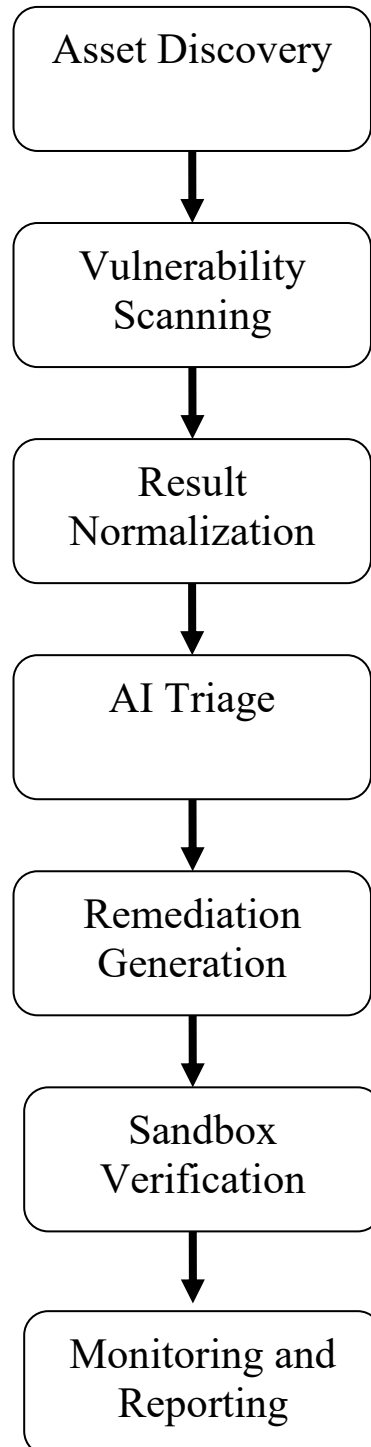
Time	Planned Work
Week 1	Problem study, requirement analysis, identification of users and security assessment needs, and finalization of project scope.
Week 2	System architecture design, module identification, data-flow design, database planning, and definition of scanner/AI workflow.
Week 3	Initial development of the Next.js frontend and FastAPI backend, including basic dashboard and API structure.
Week 4	Integration and initial testing of Nmap, OWASP ZAP, and Trivy scanning components.
Week 5	Implementation of asynchronous scan execution using Celery and Redis and connection of scanner outputs with the backend.
Week 6	Development of the AI triage workflow using LangChain and an LLM; creation of a unified vulnerability schema.
Week 7	Implementation of vulnerability prioritization, false-positive filtering, OWASP Top 10 mapping, and remediation generation.
Week 8	Development of Docker-based sandbox verification for proposed fixes and recording of verification results.
Week 9	Integration of real-time dashboard updates, live scan status, severity counters, vulnerability logs, and action controls.
Week 10	Implementation of compliance-oriented assessment and generation of audit-ready PDF/JSON reports.
Week 11	Complete system testing, debugging, integration testing, security workflow validation, and performance checks.
Week 12	Final documentation, presentation/demo preparation, refinement of the project, and compilation of the final report.



Research Methodology to be Employed:

- The project will follow a modular and iterative methodology in which individual cybersecurity components are integrated into a common automated workflow. The first stage is asset discovery and profiling, through which the platform identifies and tracks the systems or targets that require assessment. This establishes the foundation for continuous monitoring.
- In the second stage, specialized scanning agents are used for different layers of assessment. Nmap performs network discovery and port scanning, OWASP ZAP supports Dynamic Application Security Testing (DAST), and Trivy scans containers and filesystems for vulnerabilities. Their outputs may be produced in different formats; therefore, the next stage converts the raw JSON/XML results into a unified schema for centralized processing.
- The third stage is AI-driven triage. The unified findings are passed to LangChain and an LLM-based analysis workflow. The AI examines the context of findings, helps identify potential false positives, assigns meaningful priority, and maps applicable findings to the OWASP Top 10. This stage is intended to reduce the amount of manual effort required to interpret raw scanner results.
- The fourth stage is remediation assistance. Based on vulnerability details, the AI workflow generates proposed remediation actions or code patches. Rather than directly applying an unverified fix, the system sends the proposed change to a controlled Docker-based sandbox. The verification engine evaluates whether the proposed fix addresses the issue without introducing unwanted effects.
- The final stage is monitoring and reporting. Results from scans and verification are presented through the real-time dashboard using live updates. Security metrics are categorized into Critical, High, Medium, and Low severity levels. The system also maintains vulnerability logs and provides compliance-oriented reporting related to SOC2 and ISO27001

The complete methodology therefore follows the cycle:





Community Engineering Project Outcome:

- KRYNTRA is intended to provide a practical cybersecurity assessment solution that can support organizations, especially Small and Medium Enterprises, in identifying and responding to security weaknesses more continuously. By automating repetitive scanning and assisting with interpretation of results, the system can reduce dependence on entirely manual security assessment processes.
- The platform can help users understand security risks through severity-based metrics, structured vulnerability information, and a centralized dashboard. Its AI-assisted triage is intended to reduce noise from raw scanner results, while sandbox verification provides an additional safety layer before a proposed remediation is deployed. Compliance-oriented reporting can also help organizations maintain structured evidence of security findings and remediation activities.
- From an engineering and societal perspective, the project demonstrates the practical use of AI, automation, web technologies, distributed task processing, and containerization to address a real cybersecurity problem. The intended outcome is a system that helps reduce the period during which vulnerabilities remain unidentified or unaddressed and makes continuous security assessment more accessible.



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